Where Bias Comes From KEY

Name _____

Work with a partner. Compare answers when you finish each problem.

1. He or she? “O bir doktor. O bir hemşire.” (Turkish: *o* means he **or** she.)

My prediction: **doctor:** he / she **nurse:** he / she

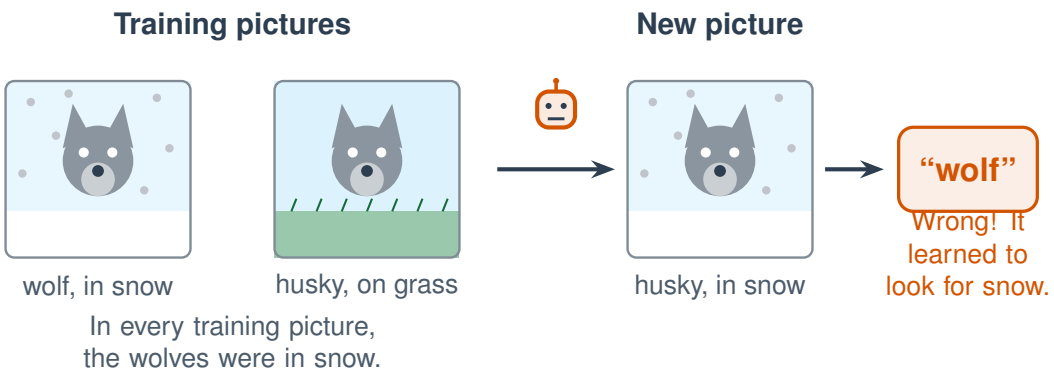
What the AI said: Answers vary. Often “He is a doctor. She is a nurse.”

Why? It copied a pattern from its training data.

2. Three ways bias gets in.

Way	Example
The data copies the past.	Amazon’s hiring AI learned from 10 years of mostly male hires.
Some people are missing.	Face programs were wrong up to 35% of the time for darker-skinned women.
Shortcuts.	The wolf AI learned to look for snow.

3. The wolf detector.



Why did the AI call the husky a wolf? In training, every wolf was in snow.

It learned to look for snow, not the animal.

4. Act it out: Clever Hans.

Round	Was Hans right?	Why?
1. The crowd knows the answer.	yes no	He watched the crowd lean back.
2. Nobody knows the answer.	yes no	No one gave him a signal.
3. Hans cannot see anyone.	yes no	He could not see the signal.

Clever Hans is like the wolf AI because both found a shortcut, not the real reason.

5. Test he and she in my language. My language: Answers vary.

English	The AI wrote	Man, woman, or both?
The doctor is busy.		
The nurse is busy.		
The engineer is tired.		
The teacher is kind.		

What I noticed: Answers vary.

6. Talk about it. The people who should fix bias in AI are Answers vary.

because _____